A FORMALLY SMOOTH LOCAL HOMOMORPHISM OF NOETHERIAN LOCAL RINGS IS FLAT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Formal smoothness is preserved by completion, completions of Noetherian rings are Noetherian, and flatness descends along faithfully flat maps. Lemma \ref{lemma-formally-smooth-completion}, Algebra, Lemma \ref{algebra-lemma-completion-Noetherian-Noetherian}, Algebra, Lemma \ref{algebra-lemma-flatness-descends-more-general}, and Algebra, Lemma \ref{algebra-lemma-completion-faithfully-flat}, cited at proof lines 9988-9992. Reduce the claim to the case of complete local rings.
- Existence of the diagram of Noetherian complete local rings with flat vertical map. Lemma \ref{lemma-embed-map-Noetherian-complete-local-rings}, cited at proof lines 9993-9998. Supplies the power series ring whose base change is a flat module mapping onto the target.
- A formally smooth algebra map whose target is a quotient of the source has a section. Lemma \ref{lemma-lift-continuous}, cited at proof lines 10001-10003. Produces the section that exhibits the target as a direct summand.
- Base change of a flat module is flat. Algebra, Lemma \ref{algebra-lemma-flat-base-change}, cited at proof lines 10007-10009. Makes the base changed power series ring flat over the source.

Source reference. Original lemma 07NP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9769–9769. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 07ED: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9443–9443. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0316: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3973–3973. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0584: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6767–6767. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00MC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 818–818. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07NN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9768–9768. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07NJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9764–9764. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00HI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 644–644. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $A \to B$ be a local homomorphism of Noetherian local rings. Assume $A \to B$ is formally smooth in the $\mathfrak{m}_B$-adic topology. Then $A \to B$ is flat.*

Proof. We may assume that A and B a Noetherian complete local rings by Lemma 07ED and Algebra, Lemma 0316 (this also uses Algebra, Lemma 0584 and 00MC to see that flatness of the map on completions implies flatness of $A \to B$). Choose a commutative diagram

$$\begin{array}{ccc} S & \longrightarrow & B \\ \uparrow & & \uparrow \\ R & \longrightarrow & A \end{array}$$

as in Lemma 07NN with $R \to S$ flat. Let $I \subset R$ be the kernel of $R \to A$. Because B is formally smooth over A we see that the A-algebra map

$$S/IS \longrightarrow B$$

has a section, see Lemma 07NJ. Hence B is a direct summand of the flat A-module S/IS (by base change of flatness, see Algebra, Lemma 00HI), whence flat. □